

1.5 Surface Area Worksheet

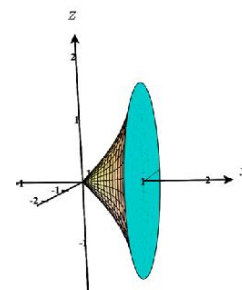
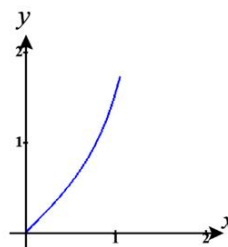
1. Setup, but do not evaluate, an integral for the surface area of the surface obtained by rotating $y = \tan(x)$ on $\left[0, \frac{\pi}{3}\right]$ about the...

- a) x -axis (setup integral with respect to x and y)

Respect to x : Note that $y = \tan(x)$ is non-negative on

$\left[0, \frac{\pi}{3}\right]$ and $y' = \sec^2(x)$ is continuous on $\left[0, \frac{\pi}{3}\right]$

Therefore, the surface area is given by $S = \int_0^{\pi/3} 2\pi \tan(x) \sqrt{1 + (\sec^2(x))^2} dx$.



Respect to y : First note that $y = \tan(x)$ is one-to-one on $[0, \pi/3]$ and this interval is within the range of the inverse tangent function, and so we can use the inverse tangent function here to solve this equation for x . $y = \tan(x) \rightarrow x = \tan^{-1}(y)$ on $0 \leq y \leq \sqrt{3}$.

Note that $x = \tan^{-1}(y)$ is non-negative on $[0, \sqrt{3}]$ and $x' = \frac{1}{1+y^2}$ is continuous on $[0, \sqrt{3}]$.

Therefore, the surface area is given by $S = \int_0^{\sqrt{3}} 2\pi y \sqrt{1 + \left(\frac{1}{1+y^2}\right)^2} dy$

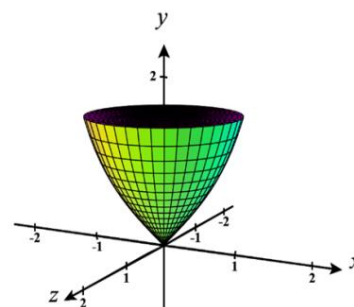
- b) y -axis (setup integrals with respect to x and y)

Respect to y : As noted above we can solve this equation for x .

$y = \tan(x) \rightarrow x = \tan^{-1}(y)$ on $0 \leq y \leq \sqrt{3}$. Note that $x = \tan^{-1}(y)$ is non-

negative on $[0, \sqrt{3}]$ and $x' = \frac{1}{1+y^2}$ is continuous on $[0, \sqrt{3}]$.

Therefore, the surface area is given by $S = \int_0^{\sqrt{3}} 2\pi \tan^{-1}(y) \sqrt{1 + \left(\frac{1}{1+y^2}\right)^2} dy$



Respect to x : Note $y = \tan(x)$ is non-negative on $[0, \pi/3]$ and $y' = \sec^2(x)$ is continuous on $[0, \pi/3]$.

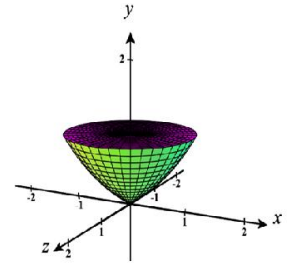
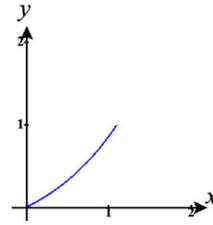
Therefore, the surface area is given by $S = \int_0^{\pi/3} 2\pi x \sqrt{1 + (\sec^2(x))^2} dx$

2. Setup, but do not evaluate, an integral for the surface area of the surface obtained by rotating $x = \ln(2y+1)$ on $0 \leq y \leq 1$ about the...

a) y -axis (create integrals with respect to x and y).

Respect to y : Note that $x = \ln(2y+1)$ is non-negative on

$[0,1]$ and $x' = \frac{2}{2y+1}$ is continuous on $[0,1]$.



Therefore, the surface area is given by
$$S = \int_0^1 2\pi \ln(2y+1) \sqrt{1 + \left(\frac{2}{2y+1}\right)^2} dy .$$

Respect to x : We first solve the given equation for y .

$$x = \ln(2y+1) \rightarrow e^x = 2y+1 \rightarrow y = \frac{1}{2}(e^x - 1) \text{ on } 0 \leq x \leq \ln(3).$$

Note that $y = \frac{1}{2}(e^x - 1)$ is non-negative on $[0, \sqrt{3}]$ and $y' = \frac{1}{2}e^x$ is continuous on $[0, \sqrt{3}]$.

Therefore, the surface area is given by
$$S = \int_0^{\ln(3)} 2\pi x \sqrt{1 + \left(\frac{1}{2}e^x\right)^2} dx .$$

b) x -axis

Respect to x : We first solve the given equation for y .

$$x = \ln(2y+1) \rightarrow y = \frac{1}{2}(e^x - 1) \text{ on } 0 \leq x \leq \ln(3)$$

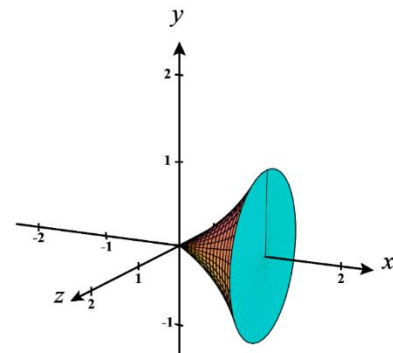
Note that $y = \frac{1}{2}(e^x - 1)$ is non-negative on $[0, \sqrt{3}]$ and $y' = \frac{1}{2}e^x$ is continuous on $[0, \sqrt{3}]$.

Therefore, the surface area is given by
$$S = \int_0^{\ln(3)} 2\left(\frac{1}{2}(e^x - 1)\right) \sqrt{1 + \left(\frac{1}{2}e^x\right)^2} dx$$

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Respect to y : Note that $x = \ln(2y+1)$ is non-negative on $[0,1]$ and $x' = \frac{2}{2y+1}$ is continuous on $[0,1]$.

Therefore, the surface area is given by
$$S = \int_0^1 2\pi y \sqrt{1 + \left(\frac{2}{2y+1}\right)^2} dy .$$

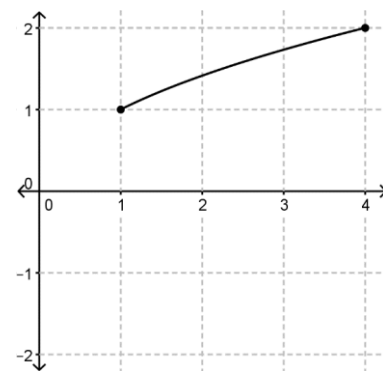


3. Find the surface area of the solid generated by revolving the curve $y = \sqrt{x}$, $1 \leq x \leq 4$ about the x -axis.

Set up integrals with respect to x and y and then evaluate one of them.

Respect to x :

We note that $y = \sqrt{x}$ is non-negative on the interval $[1, 4]$ and $y' = \frac{1}{2}x^{-\frac{1}{2}}$ is continuous on the interval $[1, 4]$.



Therefore the surface area of the solid is given by:
$$S = \int_1^4 2\pi\sqrt{x} \sqrt{1 + \left(\frac{1}{2}x^{-\frac{1}{2}}\right)^2} dx$$

We calculate the surface area to be the following.

$$S = \int_1^4 2\pi\sqrt{x} \sqrt{1 + \left(\frac{1}{2}x^{-\frac{1}{2}}\right)^2} dx = 2\pi \int_1^4 \sqrt{x} \sqrt{1 + \frac{1}{4x}} dx = 2\pi \int_1^4 \sqrt{x + \frac{1}{4}} dx = 2\pi \cdot \frac{2}{3} \left(x + \frac{1}{4}\right)^{\frac{3}{2}} \Big|_1^4 = \frac{4\pi}{3} \left[\left(\frac{17}{4}\right)^{\frac{3}{2}} - \left(\frac{5}{4}\right)^{\frac{3}{2}} \right]$$

Respect to y :

We solve the given equation for x : $y = \sqrt{x} \rightarrow x = y^2$, where y ranges from $y = 1$ to $y = 2$.

We note that $x = y^2$ is non-negative from $y = 1$ to $y = 2$ and $x' = 2y$ is continuous from $y = 1$ to $y = 2$.

Therefore the surface area of the solid is given by:
$$S = \int_1^2 2\pi y \sqrt{1 + (2y)^2} dy$$

We calculate the surface area to be the following.
$$S = \int_1^2 2\pi y \sqrt{1 + (2y)^2} dy = 2\pi \int_1^2 y \sqrt{1 + 4y^2} dy$$

Let $u = 1 + 4y^2$ this implies that $du = 8y dy \rightarrow \frac{1}{8} du = y dy$. Our bounds change to

$$u(1) = 1 + 4(1)^2 = 5 \quad \& \quad u(2) = 1 + 4(2)^2 = 17.$$

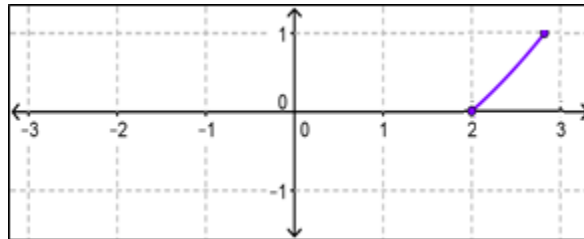
Therefore we have:

$$S = \int_1^2 2\pi y \sqrt{1 + (2y)^2} dy = 2\pi \int_1^2 y \sqrt{1 + 4y^2} dy = 2\pi \int_5^{17} \frac{1}{8} \sqrt{u} du = \frac{\pi}{4} \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_5^{17} = \frac{\pi}{6} \left[17^{\frac{3}{2}} - 5^{\frac{3}{2}} \right]$$

4. Find the surface area of the solid generated by revolving the curve $x = 2\sqrt{1+y}$, $0 \leq y \leq 1$ about the y -axis. Set up integrals with respect to x and y and then evaluate one of them.

Respect to y :

We note that $x = 2\sqrt{1+y}$ is non-negative on the for $0 \leq y \leq 1$ and $x' = (1+y)^{-\frac{1}{2}}$ is continuous for $0 \leq y \leq 1$.



Therefore the surface area of the solid is given by:
$$S = \int_0^1 2\pi \cdot 2\sqrt{1+y} \sqrt{1 + \left((1+y)^{-\frac{1}{2}}\right)^2} dy$$

We calculate the surface area to be the following.

$$\begin{aligned} S &= \int_0^1 2\pi \cdot 2\sqrt{1+y} \sqrt{1 + \left((1+y)^{-\frac{1}{2}}\right)^2} dy = 4\pi \int_0^1 \sqrt{1+y} \sqrt{1 + \frac{1}{1+y}} dy = 4\pi \int_0^1 \sqrt{1+y+1} dy = 4\pi \int_0^1 \sqrt{y+2} dy \\ &= 4\pi \cdot \frac{2}{3} (y+2)^{\frac{3}{2}} \Big|_0^1 = \frac{8\pi}{3} \left[3^{\frac{3}{2}} - 2^{\frac{3}{2}} \right] \end{aligned}$$

Respect to x :

We solve the given equation for y : $x = 2\sqrt{1+y} \rightarrow y = \frac{x^2}{4} - 1$, where $2 \leq x \leq 2\sqrt{2}$.

We note that $y = \frac{x^2}{4} - 1$ is non-negative for $2 \leq x \leq 2\sqrt{2}$ and $y' = \frac{1}{2}x$ is continuous for $2 \leq x \leq 2\sqrt{2}$.

Therefore the surface area of the solid is given by:
$$S = \int_2^{2\sqrt{2}} 2\pi x \sqrt{1 + \left(\frac{1}{2}x\right)^2} dx = 2\pi \int_2^{2\sqrt{2}} x \sqrt{1 + \frac{1}{4}x^2} dx$$

Let $u = 1 + \frac{1}{4}x^2$ this implies that $du = \frac{1}{2}x dx \rightarrow 2du = x dx$. Our bounds change to

$$u(1) = 1 + \frac{1}{4}(2)^2 = 2 \quad \& \quad u(2\sqrt{2}) = 1 + \frac{1}{4}(2\sqrt{2})^2 = 3.$$

Therefore we have:
$$S = 2\pi \int_2^{2\sqrt{2}} x \sqrt{1 + \frac{1}{4}x^2} dx = 2\pi \int_2^3 2\sqrt{u} du = 4\pi \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_2^3 = \frac{8\pi}{3} \left[3^{\frac{3}{2}} - 2^{\frac{3}{2}} \right]$$

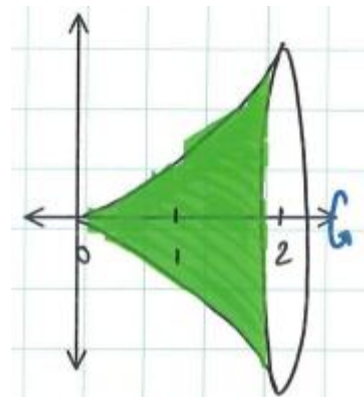
5. Consider the curve $y = \sqrt[3]{x}$ on the interval $0 \leq y \leq 2$ and the resulting surface when this curve is rotated about the y -axis. Setup and evaluate, an integral that represents the surface area of this surface.
- a) Setup and evaluate the integral with respect to y .

We first note that $y = \sqrt[3]{x} \rightarrow x = y^3$.

We see that $x = y^3$ is non-negative on $[0, 2]$ and $x' = 3y^2$ is continuous on $[0, 2]$.

Therefore, the surface area is given by

$$S = \int_0^2 2\pi y^3 \sqrt{1 + (3y^2)^2} dy = \int_0^2 2\pi y^3 \sqrt{1 + 9y^4} dy.$$



Let $u = 1 + 9y^4$ this implies that $du = 36y^3 dy \rightarrow \frac{1}{36} du = y^3 dy$. Our bounds change to

$$u(0) = 1 + 9(0)^4 = 1 \quad \& \quad u(2) = 1 + 9(2)^4 = 145.$$

Therefore we have:

$$S = \int_0^2 2\pi y^3 \sqrt{1 + (3y^2)^2} dy = 2\pi \int_0^2 y^3 \sqrt{1 + 9y^4} dy = 2\pi \int_1^{145} \frac{1}{36} \sqrt{u} du = \frac{\pi}{18} \int_1^{145} \sqrt{u} du = \frac{\pi}{18} \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_1^{145} = \frac{\pi}{27} \left[145^{\frac{3}{2}} - 1 \right]$$

- b) Setup the integral with respect to x .

We see that the portion of the curve $y = \sqrt[3]{x}$ that we are working with exists on the interval $0 \leq x \leq 8$.

Note that $y = \sqrt[3]{x}$ is continuous on $[0, 8]$, however $y' = \frac{1}{3} x^{-2/3}$ is NOT continuous on $[0, 8]$ (specifically at $x = 0$). Therefore we cannot guarantee that the surface area formula works here and we cannot continue.

6. Find the surface area of the solid generated by revolving $y = x^2$ on $[0, 2]$ about the y -axis.

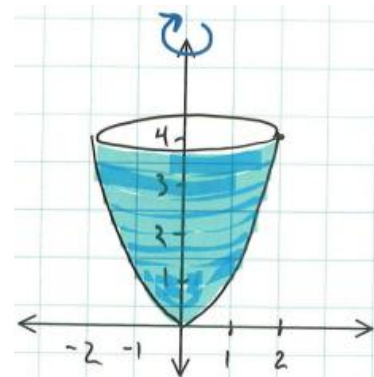
We start by solving the given equation for x . $y = x^2 \rightarrow x = \sqrt{y}$, where y ranges from 0 to 4.

We note that $x = \sqrt{y}$ is non-negative from $y = 0$ to $y = 4$ and $x' = \frac{1}{2}y^{-\frac{1}{2}}$ is NOT continuous from $y = 0$ to $y = 4$ (specifically at $y = 0$). Therefore we cannot set up this integral with respect to y . We will instead set up an integral with respect to x .

We note that $y = x^2$ is continuous on $[0, 2]$ and $y' = 2x$ is continuous on $[0, 2]$

Therefore the surface area of the solid is given by:

$$S = \int_0^2 2\pi x \sqrt{1 + (2x)^2} dx = \int_0^2 2\pi x \sqrt{1 + 4x^2} dx$$



We calculate the surface area to be the following.

$$S = \int_0^2 2\pi x \sqrt{1 + 4x^2} dx = 2\pi \int_1^{17} \frac{1}{8} \sqrt{u} du = \frac{\pi}{4} \cdot \frac{2}{3} (u)^{\frac{3}{2}} \Big|_1^{17} = \frac{\pi}{6} \left[(17)^{\frac{3}{2}} - 1 \right]$$

7. Consider the curve $y = \sin(\pi x)$ on the interval $0 \leq x \leq 1$ and the resulting surface when this curve is rotated about the x -axis. Setup, but do not evaluate, an integral that represents the surface area of this surface.

- a) Setup the integral with respect to x .

We note that $y = \sin(\pi x)$ is non-negative on $[0, 1]$ and $y' = \pi \cos(\pi x)$ is continuous on $[0, 1]$.

Therefore, the surface area is given by

$$S = \int_0^1 2\pi \sin(\pi x) \sqrt{1 + (\pi \cos(\pi x))^2} dx$$

- b) Explain why setting up this integral with respect to y would be very difficult.

In order to start integrating with respect to y we would have to solve the equation $y = \sin(\pi x)$ for x . However, on the interval $[0, 1]$ we note that $y = \sin(\pi x)$ is not one-to-one and so solving for x isn't possible with a single equation. Therefore, we would instead have to create two different equations and then create two different surface area integrals with respect to y .

